DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA
Politecnico di Milano

Machine Learning (Code: 097683)
July 20, 2021

Surname:
Name:
Student ID:
Row: Column:

Time: 2 hours Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 45 min before the end of the exam you will get 3 points, if you finish earlier than 30 min you will get 2 points and if you finish earlier than 15 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Describe the logistic regression model and how it is trained.

Exercise 2 (7 marks)

Describe the value iteration algorithm and its properties.

Exercise 3 (2 marks)

Consider the following snippet of code taking as input a dataset `X` having 100 samples with 5 features. Are the subsequent statements true or false? Motivate your answers.

```
1 import numpy as np
2 X_tilde = X - np.mean(X, axis=0)
3 C = np.dot(X_tilde.T, X_tilde)
4 eigenvalues, W = np.linalg.eig(C)
5 T = np.dot(X_tilde, W[:,:2])
```

1. The code snippet above is implementing a feature selection technique.
2. Line 2 is unnecessary if the data in `X` are scaled.
3. A model trained with the inputs `T` is likely to display a lower bias than a model trained with inputs `X`.
4. It is possible to recover the original dataset `X` from `T` by computing $X = W[:, :2]^\top \cdot T$.

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1. FALSE: the code snippet is implementing the PCA, which is a feature extraction technique.
2. FALSE: the data in `X` has to be centered anyway to compute the covariance matrix.
3. FALSE: the dataset `T` has lower dimensionality than the dataset `X`. Thus, a model trained on `T` is likely to have a larger bias but a smaller variance.
4. FALSE: in this way we are minimizing the reconstruction error, but it would be greater than zero in general.

Exercise 4 (2 marks)

We are faced with the problem of learning the correct dosage of food to give to a kitten. Each day, we want to determine the amount of food to give him/her not to make him/her starve or get too much food. At the end of each day, we know if the choice we made was below the proper dosage, if the kitten asks for more food, or if we provided him/her with more food than required, if the kitten is sleeping too much. We would like to be able to learn the proper amount while maximizing the satisfaction of the kitten over time.

1. Can the above problem be solved using a machine learning model? Which one? Provide and motivate either the features/target or action/states/reward for the chosen model.
2. Provide an algorithm to solve the above problem, commenting on the pros and cons of its usage.
3. What would change if the amount of food we provided to the cat would influence the kitten's weight over time?

1. It can be modeled as a stochastic MAB, where the reward is the satisfaction of the kitten and the action corresponds to the different amounts of food.
2. We can use both UCB1 or TS to solve this problem. If we have prior information we can exploit (previous information on other kittens) the choice of TS is advised.
3. In this case the action would influence the state of the kitten, therefore we would switch to an RL setting, therefore the use of UCB1 and TS is not a viable option anymore.

Exercise 5 (2 marks)

Tell whether the following statements about training a supervised learning model are true or false. Motivate your answers.

1. The availability of a large dataset might lead to choosing a more complex model.
2. When data is very noisy, it is not a good idea to employ regularization.
3. The larger is the training set, the smaller would be the training error.
4. Increasing the number of features would generally lead to a decrease of the training error.

1. TRUE If one has more samples, he/she might also consider more complex model to be used on the available data. Recall that the availability of a large amount of data does not mean that the model should be necessarily complex. Instead, the complexity should be appropriate to the process generating data.
2. FALSE Regularization techniques are able to avoid overfitting, therefore they are less prone to model the noise present in the data.
3. FALSE (generally) If we assume that we fix the model, as we include more samples we have that the generally the training error is increasing or remains the same. In some specific case it might happen that the error decreases.
4. TRUE We are more and more prone to overfitting as we increase the number of features, which leads to a decrease of the training error, but to lower generalization capabilities.

Exercise 6 (2 marks)

Tell whether the following statements about Kernel Methods are true or false. Motivate your answers.

1. There is no point in using a kernel dual representation, if its corresponding feature space is known.
2. $k(x, x') := e^{-\|x+x'\|^2}$ is a valid kernel function.
3. To compute prediction on a new sample using the dual representation, knowing the Gram matrix is sufficient.
4. It is possible to use Gaussian Processes in an online way (i.e., in a sequential learning setting).

1. FALSE One might use the kernel trick for computational reasons, for instance if the computation of the features in the original feature space is to computationally heavy.
2. FALSE. $k(x, x') = e^{-\|x+x'\|^2}$ cannot be used as a similarity measure. To see it, we observe that, with this function, an element is not similar to itself. Looking at its decomposition:

   $$e^{-\|x+x'\|^2} = e^{-x^\top x - x'^\top x' - 2x^\top x'} = e^{x^\top x} e^{-2x^\top x'} e^{x'^\top x'}, \tag{1}$$

   differently from the case in which we have $\|x - x'\|$, the inner term is not a valid kernel, due to the minus sign, hence, property (2) cannot be applied. These properties are, though, sufficient conditions.

   A more formal way to prove it consists in providing a counterexample: consider $\mathbb{R}$ as the feature space and consider the dataset composed by two points: 1 and -1. By computing the Gram matrix of the kernel, one can see that its determinant, which corresponds also to the eigenvalues product, is negative. Therefore at least one of them is negative and the Gram matrix is not positive semidefinite.
3. FALSE Since we need also to compute the value of the kernel between the points in the training set and the newly processed one.
4. TRUE The process requires to include a new row and a new column in the Gram matrix and to compute the kernel similarity for an additional point every time we include a new point in the training set. The process is similar to the one use for recursive least square applied to the Gram matrix.

Exercise 7 (4 marks)

A tomato factory wants to build a classifier that predicts whether tomato puree cans will be either approved (class 1) or rejected (class 0). The following binary features are evaluated: presence of organic (**ORGANIC**) or inorganic (**INORGANIC**) undesired content, a weight higher (**HIGHER**) or lower (**LOWER**) than the nominal one, and a *non-standard red* color for the puree (**PALE**). Given the following dataset:

	ORGANIC	INORGANIC	HIGHER	LOWER	PALE	**APPROVED**
Training	1	0	0	0	1	0
	0	1	0	0	0	0
	1	1	1	0	0	0
	0	0	1	0	1	0
	0	1	0	0	1	0
	0	0	1	0	0	1
	1	0	0	0	1	1
	0	0	0	1	0	1
Test	0	0	0	1	0	?
	0	1	0	0	1	?

1. Estimate a Naïve Bayes classifier from the *training samples*, choosing the proper distributions for the classes priors and the feature posteriors.
2. Predict the probability of the *test samples* to be approved or not.

1. Prior distribution for the classes modeled as Bernoulli:

$$P(c=0) = \frac{5}{8} \qquad P(c=1) = \frac{3}{8}$$

Distribution for each input given the class as Bernoulli:

$$\begin{aligned} P(x_1=1|c=0) &= \frac{3}{5} & P(x_1=0|c=0) &= \frac{2}{5} \\ P(x_1=1|c=1) &= \frac{1}{3} & P(x_1=1|c=0) &= \frac{2}{3} \\ P(x_2=1|c=0) &= \frac{3}{5} & P(x_2=0|c=0) &= \frac{2}{5} \\ P(x_2=1|c=1) &= 0 & P(x_2=0|c=1) &= 1 \\ & & &\ldots \end{aligned}$$

2. Since $P(x_4=1|c=0)=0$ the first sample has probability equal to 1 to belong to the positive class. Similarly, since $P(x_1=1|c=1)=0$ the second sample has probability equal to 1 to belong to the negative class.

Exercise 8 (4 marks)

Consider the set of trajectories below, which are obtained while running the Monte Carlo policy iteration algorithm in an MDP with three states $\mathcal{S} = \{A, B, C\}$ (A is the initial state, C is terminal), two actions $\mathcal{A} = \{u, d\}$, and discount factor $\gamma = 1$.

$$(A, u, 2) \rightarrow (B, d, -2) \rightarrow (A, d, -2) \rightarrow (A, u, 4) \rightarrow (C)$$
$$(A, d, 1) \rightarrow (B, u, -3) \rightarrow (C)$$
$$(A, d, 7) \rightarrow (B, d, 0) \rightarrow (C)$$

1. Provide the policy evaluation step according to Monte Carlo first-visit.
2. Provide the policy improvement step, i.e., the policy that the algorithm will deploy at the next iteration.
3. Do you think the results would have changed with Monte Carlo every-visit evaluation?

1. To perform the subsequent policy improvement we need to evaluate the Q function, i.e.,

$$Q(A, u) = 2 \qquad Q(A, d) = \frac{2 - 2 + 7}{3} = \frac{7}{3}$$
$$Q(B, u) = -3 \qquad Q(B, d) = \frac{0 + 0}{2} = 0$$

2. In Monte Carlo policy iteration we have to ensure sufficient exploration by deploying an ϵ-greedy policy, i.e.,

$$\pi(u|A) = \epsilon, \quad \pi(d|A) = 1 - \epsilon, \quad \pi(u|B) = \epsilon, \quad \pi(d|B) = 1 - \epsilon$$

3. The only state-action pair that occurs more than once in single trajectory is (A, u), thus we have to recompute its evaluation,

$$Q(A, u) = \frac{2 + 4}{2} = 3,$$

since $Q(A, u) > Q(A, d)$ according to every-visit evaluation, the ϵ-greedy policy would change as follows $\pi(u|A) = 1 - \epsilon$, and $\pi(d|A) = \epsilon$.